WASP-50b

R-1238

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_151224` · created 2026-07-31T10:46:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457380.781 +/- 0.052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	3.0 +/- 1.78 [%]
Orbital Inclination (inc)	89.35 +/- 2.9
Airmass coefficient 1 (a1)	1.01 +/- 0.21
Airmass coefficient 2 (a2)	-0.039 +/- 0.074
Scatter in the residuals of the lightcurve fit is	21.73 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.075 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.1404 (deviation 0.56σ)
Duration fit vs published	0.075 d vs 0.0754 d (deviation 0.01σ)
Mid-time O–C	31.3 min
Depth SNR	3.0
Residual scatter	21.73 %

Light curve

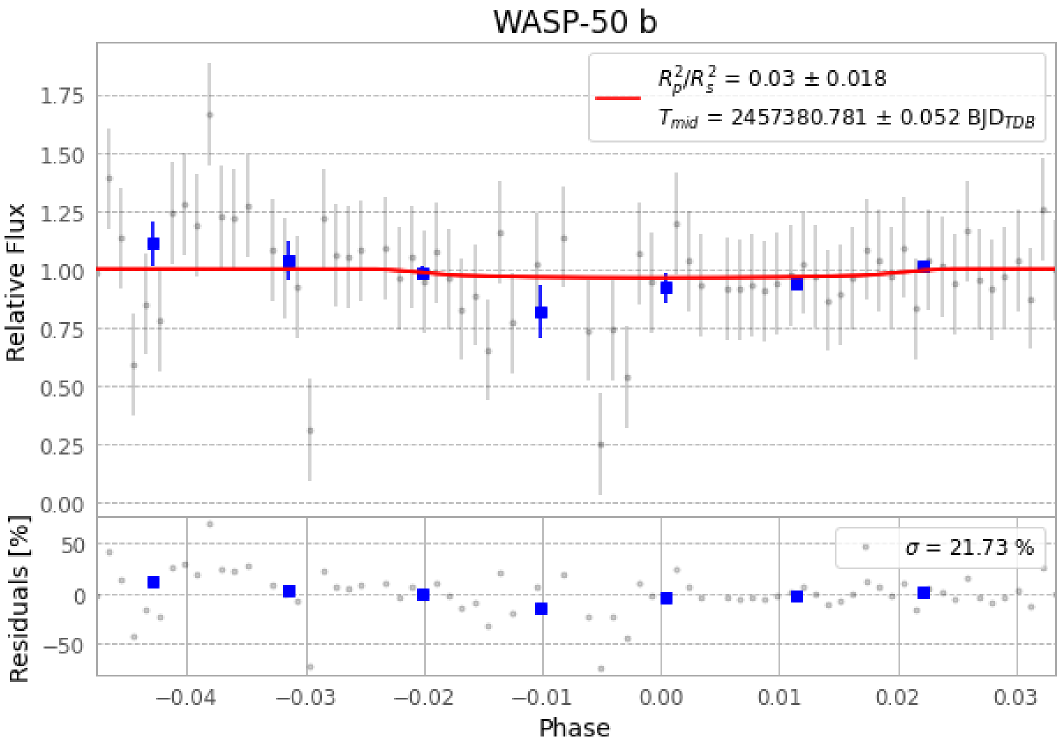

Figure 1 — FinalLightCurve_WASP-50 b_2015-12-23.png (SHA-256 cec2a10c395de6aa…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [547, 246], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6abd3aa5be44f624…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

77 FITS frames (51 MB), `WASP-50151224042412.FITS` → `WASP-50151224081212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-151224.log (f01d15e9b831…), FinalLightCurve_WASP-50 b_2015-12-23.png (cec2a10c395d…), FinalLightCurve_WASP-50 b_2015-12-23.csv (fa2c97f6207a…)

Compute resources

Wall time 325.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)